Intro to TCP/IPv4 for AI Programming

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This class teaches the basic concepts of TCP/IPv4 so students have a better idea of how to build infrastructure outside of the bounds of IaaS companies.

Why TCP/IP Matters to Coders and AI

Architecture… Architecture… Architecture…

Understanding how TCP/IPv4 works allows you to build systems outside of the bounds of IaaS providers like AWS and Azure.

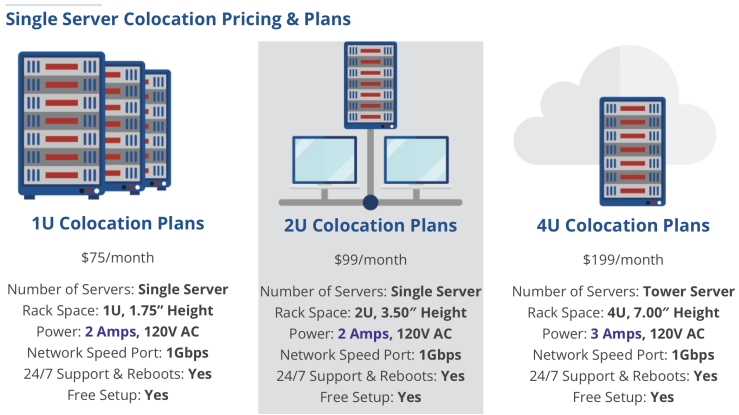
TCP/IPv4 is the language of the entire Internet and not simply a vendor's way of doing things. If you want your infrastructure to span IaaS, Virtualization, Local Bare Metal and any combination of those knowing TCP/IPv4 will allow you to build what you need.

Example:

Imagine building a YouTube clone. If you use AWS it will cost you 9 cents per 1GB of outbound traffic. A 1080p video is approx. 3GB for a one hour video so each 1 hour of video streamed will cost you $.27. 10 hours will be $2.70, 100 will be $27, 1000 will be $270 and 10,000 will be $2700. If you go "viral" the cost can grow incredibly quickly.

If you build your own server and put it in a colocation facility the upfront cost for the server may be $2,000. You can put it into a colocation rack for $200 per month and with that get 1Gbps fiber connection. A 1080p stream uses approx. 3Mbps which means that 1Gbps could theoretically support 300 simultaneous streams of 1080p. It could do this 24/7 with no additional cost.

So using AWS allows you to start very cheap, but prices rise quickly. A colocation facility requires upfront cost, but it's a set cost per month.



Example 2:

Imagine that some of the countries the users of your AI platform require data sovereignty. So their data and requests must stay with the country.

You could build a very powerful AI System for your global users that have no restrictions, but then use a colocation facility in specific countries to provide AI for their citizens.

Using Something called GeoDNS you can route users to different IP Addresses based on their location. So most users are routed to your Global AI Infrastructure, and some users are routed to systems that are local to them.

LAN, WAN, MAN, CAN, GAN, Internet

When discussing TCP/IPv4 you will hear about LAN's, WAN's and other types of networks. TCP/IPv4 is a routable protocol and so we define the network by its routers. If you have an office where all the computer connect to the Internet though a router this would be your Local Network or your LAN. If you have numerous individual offices spread that connect to each other using routers all of these together would be the WAN (Wide Area Network).

You can use other terms such as MAN, CAN and GAN based on certain factors:

- Local Area Network - You local network. Everything on your side of the router
- Wide Area Network - A network with numerous LAN's .
- Campus Area Network - A network of numerous LAN's spread over a contained geographic area
- Metropolitan Area Network - A network of numerous LAN's over a large geographic area.
- Global Area Network - A network of LAN's spread over the globe
- Internet - A specific GAN

Logical vs. Physical

When dealing with TCP/IPv4 you will need to become comfortable with the difference between "physical" and "logical" equipment and systems.

Many years ago hardware was designed to do one task, and one task only. In the modern world many times a single physical device will perform multiple functions.

So a Modem, a Router, a Switch, a Wireless Access Point, and a Firewall are all specific "logical" devices, but generally they will be built into a single SOHO "modem" that you will receive from your ISP.

When white boarding your architecture you need to build the design as if these are separate objects, and then as you solidify your design you will pick the specific products that solve your problems.

So you may have a "modem" with a wireless access point built in, but you realize it is not good enough for your situation. You would then disable the access point in your "modem" and connect/ configure an appropriate one for your network.

Many times tech professionals will use "modem" or "router" as shorthand for the device that connects the local network to the internet. This is done to ease communication for professionals, but can be confusing and is technically incorrect.

Latency

Latency is the time it takes between sending a request and receiving a response. The type of network you are on will effect latency. Generally satellite latency is higher than fiber optic. Distance also effects latency. the further two computers are a part the longer it will take the signal to travel.

When thinking about latency remember that it is additive. The time it takes for a computer to send data to a server, the time it takes the server to process that data, the time it takes for the server to pull in data from other systems such as database or REST API calls, the time it takes the server to package the data, and then the time it takes to send the data back to the requesting computer.

When planning for "quality" make sure to factor in latency. A "worse" response that is faster may be functionally better.

Network Latency

Use Case	Good	Noticeable	Bad
Web Browsing	<150 ms	150–300 ms	300+ ms
SSH/RDP	<150 ms	200 ms	300+ ms
VoIP	<150 ms	150–250 ms	250+ ms
Gaming	<80 ms	80–150 ms	150+ ms
Databases	<20 ms	50 ms	100+ ms
Industrial	<10 ms	10–20 ms	20+ ms

Beware of Caching

"Caching" is when systems save a response from another system and then provide that value when a new request is made.

Your web browser will cache web pages so that when you go back to a previously viewed page it does not have to pull all of the data from the website. Additionally systems such as DNS also cache values (Resolving Domain Names to IP Addresses). You can have problems when yuo make a change, go to test the results, but the values you get back during testing do not come from the new changes, but come from cached values.

TCP/IPv4 was designed for a different technological landscape where response times were painfully slow, and connections were not reliable. Caching improved performance under those circumstances. Now it can cause problems when you are trying to fix or modify your systems because after a change

you may get the same results. (Every tech has spent hours rewriting the same script only to find out that they were just hitting the local cache instead of their edits)

Replication is the process of data from systems communicating changes to other systems. You may here that for Internet DNS it may take 24 hours for changes to fully replicate. This means it will take up to 24 hours for your changes to be written to all the servers that provide DNS. This means that if a client communicates with one DNS server they will get the new values, but a different client may be routed to a different DNS server and get the previous ones.

OSI Model

The OSI Model is a logical model used to try to understand how network communication works. It gives us a way to talk about the individual components and processes in computer communciation.

So if someone says, "It's a Layer 1 problem" they mean that the network cable is unplugged, or damaged.

If they talk about "Layer 3" problems they are talking about IP Addresses, or IP Configuratios.

7. Application → Application User in Interacting With

6. Presentation → Computer Processing Data to Communication (Encryption)

5. Session → Maintains connection

4. Transport → TCP handles reliability

3 . Network → IP Addresses / Routers

2. Data Link → MAC Addresses / Switches

1. Physical → Physical Connections

What is a Protocol

Protocols are the languages used buy computers to talk to each other.

TCP/IPv4 is a Networking Protocol. More specifically it is a suite of Networking Protocols. TCP stands for Transmission Communication Protocol and determines how computers manage traffic between themselves. IP stands for Internet Protocol and is used for IP Addressees so that computers can find each other.

There are other protocols for routing, storage and communication.

- RIP, OSPF are Routing Protocols
- iSCSI is a Storage Protocol
- SIP, RTMP are Communication Protocols.

TCP/IPv6 ?

Tomorrow… is always a day away…

In 2000 during my MCSE NT 4.0 bootcamp I was told IPv6 would become the standard in a few years. In 2010 I created a video class laughing about IPv4 still being the predominate protocol. IPv4 is still that standard Networking Protocol that you will use.

IPv4 was supposed to die due to a lack of IP Addresses. Originally every single device connected to the Internet needed it's own IP Address and there are only approx. 4 billion IPv4 addresses possible. Something called NAT (Network Address Translation) extended the usefulness of IPv4.

Ethernet Standard

Ethernet is the network standard that most networks run on. It's important to understand that multiple Networking Protocols can run on the Ethernet standard. NWLINK, IPX/SPX, NetBIOS, Appletalk and other protocols were commonly used on Ethernet networks. TCP/IPv4 is now the general standard.

- Star Typology
- RJ45 Connectors
- MAC addresses
- CSMA/CD - Collision Domains

MAC Address (Media Access Control)

A MAC Address is Universally Unique Identifier where there should be no two identical addresses ever created. Think of it as a Social Security Number for your network card. This resides at layer 2 of the OSI model.

```
a8:51:ab:90:fd:2e
```

When Looking at a MAC Address part is the Vendor ID and part is the Serial Number. This can be used to find the vendor of a Network Card.

```
A851AB90FD2E - Apple, Inc.
```

REST API to Find Info Based on MAC Address

https://www.macvendorlookup.com/api

TCP/IPv4 - Routable Protocol Suite

TCP/IPv4 is a configurable and routable protocol suite. It allows you to build multilayer networks and provides the tools to maintain and troubleshoot them.

- •TCP (Transmission Control Protocol) - Transmission Protocol
- •IP (Internet Protocol)- Addressing Protocol
- •ICMP (Internet Control Message Protocol) - Troubleshoot communications issues (ping)
- •IGMP (Internet Group Management Protocol) - Allows for one to many broadcast communication

IPv4 (Internet Protocol)

The IP Protocol gives you the ability to give computers modifiable addresses so you can build your own routable network infrastructure.

Octets

When we discuss numbering in IP we use the Octet Numbering scheme. So the first digit is 64, the second 32, then 16,8,4,2, and finally 1. You create an octet by putting 1's over digits to add up to a number between 0 and 255

128 64 32 16 8 4 2 1

1 0 0 0 0 0 0 0 equals 128

0 0 0 1 0 1 0 1 equals 21

4 Octets are placed together seperated by periods to denote a full IP Address or Subnet Mask.

11000000.10101000.00000001.00000001 = 192.168.1.1

IP Addresses and Subnet Mask

The IP Address contains both the identifier for the Host and the identifier for the Network. The Subnet Mask is used to show where the Network Number ends and the Host Number begins.

11111111.11111111.11111111.0 = 255.255.255.0

Where the 1's are signify the Network Address.

So 192.168.1.26 with a subnet mask of 255.255.255.0 would be a 192.168.1.0 Network with a host address of 26.

Subnet masks can also be denoted with a / at the end of an IP Address. So a 192.168.1.23/24 means a subnet mask of 255.255.255.0.

When we talk about "classes" of subnets we discuss A, B and C generally. This is how many octets are used in the subnet mask.

- A = 255.0.0.0
- B = 255.255.0.0
- C = 255.255.255.0

(Scribble stuff on whiteboard about octets to impress students…)

When calculating the total number of hosts available in a network use the octet numbering to add up a total and then subtract 2. The highest number for the broadcast address, and the lowest one for the network address.

Default Gateways

Default Gateways are the Routers that the device is supposed to communicate with if an IP Address is not located on the local network. The Default Gateway will route the traffic to the WAN or INternet.

Private IP Address Blocks

Private Address Ranges are used for your private network. Networking equipment knows that these addresses should never be Internet facing and so will not try to route them on the Internet.

Your computers will communicate with the Internet through a NAT capable router. Your "external" or "Internet Facing" IP Address will be whatever your ISP Provided you, and your local IP Address will be what is configured on your computer or server.

Key Private IPv4 Address Ranges (RFC 1918)

These address blocks are reserved for private networks and will never appear on the public internet:

• Class A: 10.0.0.0 – 10.255.255.255 (10.0.0.0/8)

• Class B: 172.16.0.0 – 172.31.255.255 (172.16.0.0/12)

• Class C: 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)

Crossover Cables, Hubs and Bridges

You can use a "crossover" cable to connect 2 computers together and allow them to communicate over Ethernet and TCP/IP.

A "hub" is basically like a splitter for a cable TV cable. Communication comes in one port and then is sent out all other ports. The issue with this is that under Ethernet only one computer can talk at a time.

Ethernet uses "collision detection" so computers will communicate when they do not detect communication on the network. If 2 computers talk at the same time their communications will "collide" and this will be detected by the computers. Each computer will wait a random amount of time before trying to communicate again.

A "broadcast storm" is when too many computers are trying to send traffic, they all wait a random amount of time after a collision, but there are so many that computer keep sending traffic at the same time. There becomes a "storm" as the cycle brings all network communication to a halt.

Bridges were created to connect hubs together. They have MAC Address tables and know which side a MAC Address is on. They only forward traffic if the MAC Address is on the other side.

Switches and ARP

Switches operate on Layer 2 of the OSI Model and they route traffic based off of MAC Addresses. These are basically network bridges with multiple ports.

Switches

Switches are layer 2 networking. They use MAC addresses to identify individual computers.

When one computer tries to connect with another on a switch a "broadcast" message will be sent out to figure out which port the target computer is on. After that communication will be routed between the ports the sender and receiver are connected to.

Switches contain MAC Address Tables. These tables contain all MAC Addresses that the Switch has seen. The maximum size of these tables used to matter, but should no longer be significant.

ARP - Address Resolution Protocol

ARP Resolves MAC Addresses to IP Address. At layer 2 traffic is routed based off of MAC address and not IP Address.

When a computer needs the MAC Address of another on a TCP/IPv4 network it broadcasts a message for the IP Address it is trying to communicate with. The system with that IP Address returns its MAC Address. Then the sender will send traffic based on the MAC Address and the Switch will route it appropriately.

Example:

Run the command **arp -a** at the command line and you shoudl receieve a response like this.

```
eli@osx ~ % arp -a
? (169.254.1.0) at (incomplete) on en0 [ethernet]
? (192.168.1.79) at cc:3b:fb:4:8:b7 on en0 ifscope [ethernet]
? (192.168.1.80) at 64:9a:63:a5:ca:db on en0 ifscope [ethernet]
? (192.168.1.94) at 7a:da:1d:4f:ea:5e on en0 ifscope permanent [ethernet]
? (192.168.1.131) at 1a:53:41:57:77:2f on en0 ifscope [ethernet]
? (192.168.1.135) at d4:90:9c:d4:c5:68 on en0 ifscope [ethernet]
? (192.168.1.255) at ff:ff:ff:ff:ff:ff on en0 ifscope [ethernet]
mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
```

TCP (Transmission Control Protocol)

TCP controls the "sessions" between 2 computers. A session is the communication stream between devices and will entail things such as the TCP Port number and the 3 Way handshake.

3 Way Handshake

- SYN: Client sends a synchronization packet to the server to initiate the session.
- SYN-ACK: Server responds with a synchronization-acknowledgment packet.
- ACK: Client sends an acknowledgment packet to confirm, establishing the connection.

Windowing

Windowing is the process whereby the recieving computer can relay how stable the network connection in and ask for more data per needing an acknowledgement.

Sessions

Session configurations are set in the server software itself. This may include how many sessions can be open at a time, speed of network communication and other things.

TCP Ports

If you think of an IP Address as the street address for a building, then TCP Port numbers would be the door numbers. Each service on a computer that is used to communicate with other computers will use a specific port.

•192.168.1.1:8080

Every Protocol uses a TCP Port. They are generally preconfigured, but can be manually set.

•SMTP - 25

•HTTP - 80

•HTTPS - 443

•FTP - 20

•SSH - 22

Routers and Default Gateways

Routers connect networks together in TCP/IPv4. A Default Gateway is used when a computer cannot cannot be found on the LAN communication is sent to the Default Gateway to determine if the computer can be found outside of the LAN.

Modems

Modems change Network Types.

- Cable Modem -> Ethernet
- Fiber Optic Modem -> Ethernet
- DSL - > Ethernet

NAT (Network Address Translation) and Port Forwarding

NAT or Network Address Translation allows numerous computers to share one Public IP Address. A NAT enabled router will track which LAN computers are talking with which outside systems and then route the data appropriately.

When using NAT computers TCP Ports are not directly exposed to the Internet. Port Forwarding is a configuration where when incoming traffic reaches the router from the Internet the router will then Forward traffic to specific IP Addresses or Host Names within the LAN based on the TCP Port.

Generally one TCP Port can only be forwarded to a single computer.

Ext.Port			Protocol TCP	Protocol UDP	IP Address		Enable
80	To	80	☑	☐	192.168.1.	100	☑
0	To	0	☐	☐	192.168.1.	0	☐

Carrier Grade NAT

Some ISP's use NAT for their own ISP users. Web traffic goes through the ISP's NAT enabled routers, then through your NAT enabled router/ modem to get to your computer. In this case you will not be able to forward TCP Ports because they would have to also be forwarded from the ISP's router.

Internet Facing Static IP Addresses

You can connect a server/ computer directly to an Internet Facing IP Address and expose all of its TCP Ports. If you rent a VPS (Virtual Private Server) from companies such as DigitalOcean or Linode those servers will be directly connected to the Internet.

Firewalls

Firewalls block traffic to or from TCP Ports on a computer. You can set whether inbound or outbound originating traffic should be blocked, and which IP Addresses or Ranges the firewall should allow or block.

Firewalls can be either hardware or software. Generally a network hardware firewall will be between your LAN and the internet. You can then setup software firewalls on individual systems.

So you could setup your MySQL server to only accept traffic on port 3306 from the IP Address of your web server. Your web server would then be setup to accept traffic from anyone on port 80. This way you don't have a rogue/hacked server trying to connect to your Database server. (3306 is MySQL's default port. Port 80 is for HTTP/Web traffic)

BE CAREFUL configuring Software Firewalls on test systems. Verify your setup works BEFORE implementing firewalls

DNS (Domain Name Service)

DNS Resolves Fully Qualified Domain Names (FQDN's) to IP Addresses.

- SERVER -> 192.168.1.10
- cnn.com -> 151.101.3.5

Resolution Order

- DNS Cache - Your compuetr will cache DNS requests. (use dns flush command to clear)
- Hosts File - Text File on your local computer
- Local DNS - Either your SOHO Router/Modem, a Windows Server, or Pi Hole
- Remote DNS (ISP, CloudFlare, Google) - Recursive DNS Resolver

Reverse DNS

Reverse DNS resolves IP Addresses to FQDN's. This is done for security incase DNS was poisoned. Your computer resolves the FQDN to the IP Address, and then resolves the IP Address to the FQDN to verify that they match.

GeoDNS

Some DNS Services allow for providing DNS responses based on the geographic area the requestor is in. So cnn.com would resolve to one address for US users, and a different one for Indian ones.

This can be useful for latency issues, or for data soverignty where users data is not supposed to leave the confines of a country.

SSL (Secure Socket Layer)

SSL Encrypts traffic between 2 computers. SSL requires a server have a "certificate". Through encryption this certificate proves the server is who it says it is.

Many web functions and services require SSL/ HTTPS to function. Things from JavaScript HTML 5 API's to using Web Hooks in ElevenLabs.

You can "purchase" a certificate from your web host or DNS registrar. You can also install and configure Let's Encrypt to create certificates.

CloudFlare DNS for SSL

If you purchase your DNS Name through CloudFlare they offer a simple SSL service. You point your domain name to your servers IP address. CloudFlare acts as a proxy between your server and the visitors traffic. This way CloudFlare can provide SSL between the visitor and the CloudFlare proxy. This provides SSL functionality with little setup and no additional fees.

DHCP (Dynamic Host Configuration Protocol)

DHCP allows for IP Address to be dynamically assigned to computers on a network. Most DHCP Servers will be built into your "modem", but you can run servers to provide this service. Windows Servers have DHCP built in and tightly integrate it to security and Active Directory. Servers such as Pi Holes can offer DHCP along with DNS too.

The DHCP Server will provide all IP Configuration information. This will be the IP Address, DNS Servers, and Default Gateways.

DHCP Servers will assign IP Addresses based on a Scope that they are provided. They can be configured to offer IP Addresses between 192.168.1.100 - 192.168.1.200. This allows you to be able to assign static addresses with a certain range and not have DHCP try to assign those addresses.

Some DHCP Servers will allow for configurations for Reserved Addresses. This is where you can add addresses for servers, networking equipment, and anything else that should have a static address into the Servers settings. If you can't reserve addresses you can just setup static addresses within a range, and then make sure the DHCP scope assigns addresses outside of it.

IP Addresses are "Leased" to clients for a set amount of time that may or may not be able to be changed on the DHCP server. In a network such as an Internet Cafe or if you have employees in a hybrid work from home arrangement you may want a short lease of 24 hours so that IP Addresses are quickly recycled. In a static environment where computers are not expected to move you may want a lease of a week.

The client computer will periodically try to renew it's lease before it has expired. If it's not able to renew the lease by the time it expires then it may stop communicating on the network.

Client computers may retain the DHCP address from another network so you might have to run a release/ renew command to release the current IP Address and get a new one.

In an enterprise network DHCP and DNS should be tightly integrated, but technically they do not have to be. If they are integrated when a Host gets a new IP Address DNS will also be able to update the Name to IP Address record.

When pointing systems at each other generally use FQDN's (Fully Qualified Domain Names) instead of IP Addresses. If you hard code an IP Address and then the computer changes its address your system will break. If you use an FQDN then it will simply point to the new IP Address.

VPN

VPN's or Virtual Private Networks allow devices outside of a network to connect to the network and logically seem as if they are inside of it. They can access local printers, or resources as if they were sitting in the office.

VPN's are made up of 2 services. It creates a secure "tunnel" between the device and the network. This tunnel is used to determine if the session is trying to be hacked or compromised. If a problem is detected the tunnel is dropped, and then recreated. The concept is that since there are any number of routes to a network over the internet then by dropping and rebuilding the tunnel a new route will be used.

Inside the tunnel the traffic is encrypted using SSL. This creates a secure connection to the LAN.

VPN's can also be used between entire networks. VPN capable routers can connect to each other to create one logical network with offices that are geographically dispursed.

VPN's themselves are considered secure, but is a system is compromised then it will have access to the entire network it is connected to.

Imagine OpenClaw on a system where the user is logged in with Global Admin privileges and the user VPN's into the corporate network.

Proxy Servers

Proxy Servers were the standard way to share an internet connection before NAT. They allow numerous computers to use a single server for internet traffic. The value is that you add layers of security between computers and the internet. Also you can create a "Caching Proxy Server" that will cache requests of all traffic and be able to reuse downloaded content for other computers.

We used a SQUID Caching Proxy Server at my computer repair company. When the first computer downloaded a Microsoft Service Pack it would be downloaded from the remote servers. When the next computer needed the service pack it was delivered from the local cache.

Reverse Proxy Servers

Reverse Proxy Servers direct incoming traffic to one of numerous internal servers. This can be used for load balancing traffic to web servers, or to direct traffic to specific servers based on programmed criteria.

NGINX is a reverse proxy server that is used to serve Python Web Applications.

TCP/IP Tools

WhatIsMyIp.com

If you go to this website it will give you your external IP Address.

ping

ping allows you to see if your computer can "see" another. It will show connectivity, latency, and if you ping using a Domain Name it will return the IP Address.

```
ping -c 1 IP/DOMAINNAME
```

```
eli@osx ~ % ping -c 1 cnn.com
PING cnn.com (151.101.67.5): 56 data bytes
64 bytes from 151.101.67.5: icmp_seq=0 ttl=57 time=29.040 ms

--- cnn.com ping statistics ---
1 packets transmitted, 1 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 29.040/29.040/29.040/0.000 ms
```

traceroute

Trace Route will allow you to get responses from networking equipment between your system and the one you are trying to communicate with. ICMP may be truned off on many pieces of equipment in the path, but if it is turned on you will get a response from them.

```
traceroute IP/DOMAINNAME
```

```
eli@osx ~ % traceroute silicondojo.com
traceroute to silicondojo.com (185.230.63.107), 64 hops max, 40 byte packets
 1  192.168.1.254 (192.168.1.254)  6.526 ms  3.226 ms  2.974 ms
 2  107-128-240-1.lightspeed.gnvlsc.sbcglobal.net (107.128.240.1)  6.016 ms  5.990
ms  3.988 ms
 3  99.5.25.76 (99.5.25.76)  4.079 ms  7.846 ms  7.874 ms
 4  * * *
 5  * * *
 6  * * *
 7  * atl-b24-link.ip.twelve99.net (62.115.146.216)  20.436 ms *
 8  atl-bb1-link.ip.twelve99.net (62.115.134.246)  17.072 ms  15.126 ms  15.065 ms
 9  rest-bb1-link.ip.twelve99.net (62.115.138.70)  31.205 ms  29.126 ms  31.281 ms
10  ash-b2-link.ip.twelve99.net (62.115.138.190)  33.900 ms
    ash-b2-link.ip.twelve99.net (62.115.123.123)  31.788 ms  32.228 ms
11  wixcom-ic-341291.ip.twelve99-cust.net (62.115.169.157)  33.056 ms  35.793 ms
33.973 ms
12  * * *
13  unalocated.63.wixsite.com (185.230.63.107)  33.197 ms  28.061 ms  27.984 ms
eli@osx ~ %
```

arp -a

arp -a command can show you what is in the computers ARP table. This is the computers that your system knows about and what their MAC Addresses are.

```
arp -a
```

```
eli@osx ~ % arp -a
? (169.254.1.0) at (incomplete) on en0 [ethernet]
? (192.168.1.77) at 2e:94:1:c0:90:d on en0 ifscope [ethernet]
? (192.168.1.79) at cc:3b:fb:4:8:b7 on en0 ifscope [ethernet]
? (192.168.1.80) at 64:9a:63:a5:ca:db on en0 ifscope [ethernet]
? (192.168.1.94) at 7a:da:1d:4f:ea:5e on en0 ifscope permanent [ethernet]
? (192.168.1.131) at 1a:53:41:57:77:2f on en0 ifscope [ethernet]
? (192.168.1.132) at 4:e4:b6:59:18:70 on en0 ifscope [ethernet]
? (192.168.1.135) at d4:90:9c:d4:c5:68 on en0 ifscope [ethernet]
? (192.168.1.146) at cc:3b:fb:0:9e:b5 on en0 ifscope [ethernet]
? (192.168.1.161) at 2e:94:1:c0:90:d on en0 ifscope [ethernet]
? (192.168.1.196) at (incomplete) on en0 ifscope [ethernet]
? (192.168.1.232) at 9c:76:13:59:af:dc on en0 ifscope [ethernet]
? (192.168.1.233) at 9c:76:13:59:b7:d5 on en0 ifscope [ethernet]
? (192.168.1.245) at 76:b9:e9:16:db:6 on en0 ifscope [ethernet]
? (192.168.1.254) at a8:40:f8:82:1b:d1 on en0 ifscope [ethernet]
? (192.168.1.255) at ff:ff:ff:ff:ff:ff on en0 ifscope [ethernet]
mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
```

LAB - AI and ARP

This lab allows you to ask an LLM questions about your network by pulling in the results from the arp -a command. With the ARP data you can learn about numbers of computers, IP Addresses and Vendors.

```
Install Ollama from Ollamal.com
ollama pull gpt-oss

python3 -m pip install ollama
```

tcpip-mac-ai.py

```
from ollama import chat, ChatResponse
import os

def ai(query, network):
    query = f'''Based on this information from an arp -a command - {network}
                What is the answer to this question: {query}'''
    response: ChatResponse = chat(model='gpt-oss', messages=[
        {
        'role': 'user',
        'content': query,
        },
    ])
    return response.message.content

network = os.popen('arp -a').read()

print(network)

while True:
    query = input('Question: ')
    response = ai(query, network)
    print(query)
    print(response)
```

```
Question: what ip address does the Xiaomi device have
what ip address does the Xiaomi device have
**Answer:**
The Xiaomi device(s) in your network appear with the MAC-vendor prefix
**CC:3B:FB**. In the ARP table you posted that prefix is found on two entries:

| IP address      | MAC address          |
|-----------------|----------------------|
| **192.168.1.79** | cc:3b:fb:04:08:b7 |
| **192.168.1.146**| cc:3b:fb:00:9e:b5 |
If you only have one Xiaomi device on the network, the one with the IP
**192.168.1.79** is the most likely candidate (this is the common address for a
Mi-Router or a Mi-Smart-Plug).
If you have two Xiaomi devices, the other one will be at **192.168.1.146**.

So, look for 192.168.1.79 (and 192.168.1.146 if you need the second one).
```

LAB - Ping and AI

In this lab we will ping 3 domain names and send the results to our LLM. We can then ask the LLM questions about the results. So on my computer I modified the HOST file to point tacobell.com to 127.0.0.1 during an old test. I keep forgetting that I did that, and still haven't changed it. Ths script can tell me about the problem and any other that it sees.

tcpip-ping-ai.py

```
from ollama import chat, ChatResponse
import os

def ai(query, network):
    query = f'''Based on this information from a ping command - {network}
                What is the answer to this question: {query}'''
    response: ChatResponse = chat(model='gpt-oss', messages=[
        {
        'role': 'user',
        'content': query,
        },
    ])
    return response.message.content

host = ['cnn.com','fox.com','tacobell.com']

network = ''
for site in host:
    response = os.popen(f'ping -c 1 {site}').read()
    network+=response

while True:
    query = input('Question: ')
    response = ai(query, network)
    print(query)
    print(response)
```

```
Question: are there any obvious probems?
are there any obvious probems?
**Short answer:**
Nothing is wrong with the network itself – there’s no packet loss and the
round-trip times are perfectly fine. The only “odd” thing is that `tacobell.com` is
resolving to **127.0.0.1** (your local machine) instead of an external IP. That’s
usually the result of a deliberate hosts-file override, a local DNS entry, or a
mis-configuration.

---

### What the ping output tells us

| Host | IP resolved | RTT (ms) | TTL | Comment |
|------|-------------|----------|-----|---------|
```

```
| cnn.com | 151.101.195.5 | 29.007 | 57 | Typical for a CDN edge; low loss. |
| fox.com | 13.248.160.137 | 15.421 | 244 | TTL 244 is unusually high, but can
happen if the upstream router sets a high initial TTL; still no loss. |
| tacobell.com | 127.0.0.1 | 0.079 | 64 | This is **localhost** – pinging yourself.
No loss, negligible latency. |

- **No packet loss** – all three pings succeeded.
- **Stddev = nan** – with only one ping, standard deviation isn’t defined; that's
expected.
- **TTL values** are just the remaining hops; their differences simply reflect
different routes.

---

### The only potential issue: `tacobell.com → 127.0.0.1`

1. **Intended redirection**
   * Some environments use a hosts file entry to block or redirect a site to
localhost (e.g., a parental-control or ad-blocking setup).

2. **Mis-configuration**
   * If you didn’t set that yourself, it could indicate that someone (or some
software) has modified your DNS or hosts file.
   * Check `/etc/hosts` (Linux/macOS) or `C:\Windows\System32\drivers\etc\hosts`
(Windows) for an entry like:
     ```
     127.0.0.1   tacobell.com
     ```

3. **Security concern**
   * If the redirect appears out of nowhere, consider whether your machine has been
tampered with (malware, compromised hosts file, etc.).
   * Run a quick hosts-file audit and, if needed, run a malware scan.

---
```